Field Hours

CSCE 649 Physically-Based Modeling

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Position, velocity, and a step

You represent a bit of matter as a particle with a position, a velocity, and a mass. One particle starts at position 0 with velocity 2 and mass 1, and no force acts, so acceleration is 0. Over a time step of 0.5 you update velocity first, then position from the new velocity. Velocity stays 2. Position becomes 0 plus 2 times 0.5, which is 1. That two-update order is the integrator.

You add a constant acceleration of -10, a stand-in for gravity, and keep the step at 0.5. Velocity changes first: 2 plus -10 times 0.5 equals -3. Position then uses the new velocity: 0 plus -3 times 0.5 equals -1.5. You do not add acceleration directly onto position. The step 0.5 is large so the overshoot is obvious. A step of 0.1 would move less far under the same acceleration.

Acceleration changes velocity. The new velocity changes position. Then you repeat.

1. Acceleration from forces
2. Update the velocity
3. Update the position
4. Repeat the step

One step, two accelerations

1. Start at position 0, velocity 2, mass 1, and time step 0.5.
2. With acceleration 0, velocity stays 2 and position becomes 1.
3. With acceleration -10, velocity becomes -3 and position becomes -1.5.

Problem 1. Acceleration is 0, velocity is 2, the time step is 0.5, and position starts at 0. Where is the particle after one step?

Answer. At position 1. Velocity stays 2, and 2 times 0.5 is 1.

Problem 2. Acceleration is -10, the time step is 0.5, and velocity starts at 2. What is the velocity after the velocity update?

Answer. -3. You add -10 times 0.5, which is -5, to the starting velocity 2.

Lecture 2 · 90 minutes

Force, mass, and the floor

You compute acceleration as net force divided by mass. Two forces, 4 and -1, act on a particle of mass 2. The net

force is 3, so the acceleration is 1.5. You feed that acceleration into the velocity update. The same net force on mass 1 would accelerate at 3, twice as hard. You do not skip the division and treat the force number as an acceleration. Mass is part of the mechanism.

You answer a hit on the ground plane y = 0 with a projection, not with hope that the next step stays legal. If the updated position is y = -0.2, the particle has passed the plane by 0.2. You put it back at y = 0 and reverse the inward velocity. An inward velocity of -3 becomes 3 if the bounce coefficient is 1. With coefficient 0 the stored velocity is 0 and the particle rests on the plane.

Forces become acceleration through mass. The floor is a projection after the step.

Force to acceleration

Sum the forces first

Divide by the mass

Mass 2 and force 3

Ground at y = 0

Move y back to 0

Reverse inward velocity

Scale by the bounce

A step into the floor

1. Net force is 3 and mass is 2, so acceleration is 1.5. That value enters the velocity update.

2. A later position lands at y = -0.2. You move the particle to y = 0.

3. Inward velocity -3 with bounce coefficient 1 becomes 3. With coefficient 0 it becomes 0.

Problem 1. Forces are 4 and -1, and the mass is 2. What is the acceleration?

Answer. 1.5. The net force is 3, and 3 divided by 2 is 1.5.

Problem 2. The particle reaches y = -0.2 with velocity -3, and the bounce coefficient is 0. What do you store?

Answer. Position y = 0 and velocity 0. You place the particle on the plane and scale the reversed velocity by 0.

Lecture 3 · 90 minutes

A spring of rest length 1

You model a short flexible rod as two particles and one spring. The rest length is 1. Particle A is at 0 and particle B is at 1.6, so the stretch is 0.6. Stiffness 10 pulls with magnitude 10 times 0.6, which is 6, along the rod. You apply 6 to each particle, in opposite directions, toward the other. The pair of forces adds to 0, so the spring does not invent a net shove of the whole rod.

You damp the spring, or the integrator will keep the oscillation going. Damping opposes the rate of change of

length. If the length grows at speed 0.4 and the damping coefficient is 5, the damping magnitude is 2. You add it on the same line as the spring, in the direction that slows the growth. Without it, the energy in the stretch of 0.6 trades between the spring and the velocities on every step of size 0.5.

Spring force uses stretch. Damping uses how fast the length is changing.

1. Compare length to rest

2. Force is k times stretch

3. Damping opposes the rate

4. Push both particles

Stretch of 0.6

1. Rest length is 1, current length is 1.6, and stiffness is 10. The spring magnitude is 6.

2. A is pulled toward B and B is pulled toward A, each with magnitude 6.

3. The length grows at 0.4 and the damping coefficient is 5, so you add a damping magnitude of 2 against that growth.

Problem 1. Rest length is 1, current length is 1.6, and stiffness is 10. What is the spring force magnitude on each particle?

Answer. 6. The stretch is 0.6, and 10 times 0.6 is 6. The two forces point toward each other.

Problem 2. The length grows at speed 0.4 and the damping coefficient is 5. What damping magnitude do you add?

Answer. 2. You multiply 5 by 0.4, and that force opposes the growth of the length.

Lecture 4 · 90 minutes

Two links and two angles

You pose an arm from joint angles when the motion is prescribed rather than forced. Both links have length 2. The shoulder angle is 0 degrees, along x, and the elbow angle is 90 degrees. The elbow sits at (2, 0). The wrist sits at (2, 2), because the second link points up. You compute these positions from the angles. You do not integrate them from forces. The map from angles to positions is the kinematic mechanism.

You also get a wrist speed from the angle rates. If the elbow angle increases at 1 radian per second and the shoulder is still, the wrist moves perpendicular to the second link. That link points along y, so the motion is along x. The speed is the link length times the rate, 2 times 1, which is 2. You use this when a later constraint cares how fast the wrist moves, not only where it sits.

Angles determine positions. Angle rates determine the wrist speed.

Two joint angles -> Elbow at (2, 0) -> Wrist at (2, 2) -> Speed from the rate

The right-angle elbow

1. Both links have length 2. The shoulder angle is 0, so the elbow is at (2, 0).

2. The elbow angle is 90 degrees, so the wrist is at (2, 2).

3. The elbow rate is 1 radian per second and the shoulder rate is 0. The wrist speed is 2, perpendicular to the second link.

Problem 1. Both links have length 2, the shoulder angle is 0, and the elbow angle is 90 degrees. Where is the wrist?

Answer. At (2, 2). The elbow is at (2, 0), and the second link points up by its length of 2.

Problem 2. The elbow rate is 1 radian per second, the shoulder is still, and the second link has length 2. What is the wrist speed?

Answer. 2. Speed is link length times angular rate, and the direction is perpendicular to the second link.

Lecture 5 · 90 minutes

A number that must be 0

You state a constraint as a number that must stay 0. A particle stuck to the ground has constraint value y, which must be 0. A bead that must stay 1 unit from the origin has constraint value x squared plus y squared minus 1. At (1, 0) that value is 0, so the bead is legal. At (1.2, 0) the value is 1.44 minus 1, which is 0.44. The bead has drifted off the circle.

You remove the drift on this step instead of only attaching a spring. A spring would leave the 0.44 error and pull over time. For the bead you slide the position along the radius back to length 1, from (1.2, 0) to (1, 0). You then strip the outward part of the velocity so the next step of size 0.5 does not rebuild the same drift. The projection is the constraint mechanism.

A spring tolerates an error for a while. A projection removes it on this step.

Spring pull

Error of 0.44 remains

Force acts over time

Drift can persist

Projection

Move back to length 1

From (1.2, 0) to (1, 0)

Strip outward velocity

A bead outside the circle

1. The legal circle has radius 1. The bead is at (1.2, 0).

2. The constraint value is 1.44 minus 1, which is 0.44.

3. You project the bead to (1, 0) and remove the outward part of the velocity before the next step.

Problem 1. The bead is at (1, 0) and the required radius is 1. What is the constraint value?

Answer. 0. One squared plus zero squared minus 1 is 0, so the bead is legal and you do not move it.

Problem 2. The bead is at (1.2, 0). What position do you project it to?

Answer. (1, 0). You keep the direction and set the distance to 1. You also remove the outward velocity so the next step does not rebuild the 0.44 error.

Lecture 6 · 90 minutes

Pinning the wrist

You pin the wrist by changing joint angles, not link lengths. The wrist is at (2, 2) and the target is (2, 1), so the error vector is (0, -1) and the error length is 1. Both links stay length 2. You take a small elbow step that reduces the error. An elbow rate moves the wrist perpendicular to the second link. You do not stretch a link to close the gap.

You stop when the error length drops under 0.1. If one step cuts the error from 1 to 0.4, you step again. If a step would push the elbow past 150 degrees, you stop at 150 and say the rest is out of reach. Two links of length 2 can extend to 4, so (2, 1) may be reachable, but the angle limit can still block a step. You check the limit first.

The wrist follows the angles. The angles move only until the error or the limit stops them.

1. Wrist from the angles

2. Error against the target

3. Step the elbow angle

4. Stop under 0.1

Target at (2, 1)

1. The current wrist is (2, 2). The target is (2, 1). The error length is 1.

2. An elbow step moves the wrist perpendicular to the second link. You keep a step that reduces the error.

3. You repeat until the error length is under 0.1, and you refuse an elbow angle past 150 degrees.

Problem 1. The wrist is at (2, 2) and the target is (2, 1). What is the error length?

Answer. 1. The error vector is (0, -1), and the length of that vector is 1.

Problem 2. A proposed step would set the elbow to 160 degrees, and the limit is 150. What do you store?

Answer. 150 degrees, and you report that the step cannot be taken in full. Both links stay length 2. You do not stretch a link to buy the extra 10 degrees.

Lecture 7 · 90 minutes

Step size and order

You choose the time step from the stiff spring, not from the frame rate. Stiffness 10 on a stretch of 0.6 makes a force of 6. On mass 1, a step of 0.5 changes velocity by 3. A step of 0.1 changes it by 0.6. The smaller change is less likely to fling the particle past rest length 1. The mechanism stays the same. Only the step size changes the throw.

You still apply collisions and constraints after the force update. The order is: spring and other forces, velocity, position, then the ground, then a bead or wrist projection. If you project the bead to radius 1 and afterward evaluate the spring at the old illegal position, you put the 0.44 error back. A step of 0.1 does not repair that swapped order. You write the order next to the step and use it on every substep.

Forces first, motion second, projections last. A smaller step does not reorder them.

Spring and forces -> Update velocity -> Update position -> Project constraints

The same spring, two steps

1. The spring force magnitude is 6 and the mass is 1.

2. A step of 0.5 changes velocity by 3. A step of 0.1 changes velocity by 0.6.

3. Either way you project constraints after the position update. Projecting earlier lets the spring read a stale position.

Problem 1. The spring force magnitude is 6, the mass is 1, and the time step is 0.1. How much does that force change the velocity?

Answer. By 0.6. Force divided by mass is 6, and 6 times the step 0.1 is 0.6.

Problem 2. Why does a smaller step fail to fix a solver that projects the bead before the spring reads the position?

Answer. The spring still reads the illegal position and can reintroduce the 0.44 drift. Projection has to follow the position update. Step size does not repair a swapped order.

Lecture 8 · 90 minutes

One short shot

You can now step a particle from force and a time step, bounce it off y = 0, tie two particles with a spring of rest length 1, pose a two-link arm of lengths 2 and 2, and pin the wrist until the error is under 0.1. You can also tell a kinematic pose from a dynamic step. Angles map straight to positions. Forces reach positions only through the velocity update and the step. That split is the modeling choice.

You build one short shot in the order from the stability lecture. Use the arm to place the wrist near (2, 1), then a constraint to finish under error 0.1, without passing 150 degrees. Hang a vertical rod from that wrist on the spring. Step at 0.1, then project the ground. A frame that skips the projection can show a negative y and still be called contact. You do not accept that frame.

Pose, then constrain, then step the spring, then the floor.

1. Pose the two-link arm

2. Project the wrist error

3. Step the spring at 0.1

4. Then the ground plane

A rod hung from the wrist

1. The arm places the wrist near (2, 1) with links of length 2. The constraint finishes with error under 0.1.

2. The rod rest length is 1. At length 1.6 and stiffness 10 the spring magnitude is 6.

3. You step at 0.1, so that force changes a unit-mass velocity by 0.6, then you project any y below 0 back to the plane.

Problem 1. Combine the spring and the ground on a vertical rod. Stiffness 10, stretch 0.6, mass 1, and step 0.1 change velocity by 0.6 during the force update. The position update then lands at y = -0.2 with velocity -3, and the bounce coefficient is 0. What do you store?

Answer. y = 0 and velocity 0. The spring already acted in the force update. The ground projection then places the particle on the plane and, with bounce coefficient 0, stores velocity 0.

Problem 2. Combine the arm and the circle-style constraint. The wrist is at (2, 2), the target is (2, 1), and the elbow limit is 150 degrees. What is the error length, and what do you refuse to change?

Answer. The error length is 1. You change joint angles to reduce it. You refuse to push the elbow past 150 degrees, and you refuse to change either link length from 2.